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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

Master of Science in Computer Science
(Advanced Software Development - Full-Time)

Master of Science in Computer Science
(Data Science - Part-Time)

Year 1

SEMESTER 1
EXAMINATIONS 2023/24

DATA MINING

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Two Hours

INSTRUCTIONS TO CANDIDATES

Answer all parts in Question 1 and 2. And in question 3 answer 4 out of 5 parts.

1. Answer the following questions: (30 marks)

(a) Discuss the difference between Data Mining and Machine Learning.

- **Data Mining is discovering patterns and relationships in large datasets, it refers to the whole process. (5 marks)**
- **Machine Learning is a subset/step of data mining that focuses on training models to make predictions or decisions. (5 marks)**

(10 marks)

(b) Discuss the distinction between interpretability and explainability in the context of Machine Learning models and their connection to model performance.

- **Interpretability refers to understanding how a model makes predictions, while explainability involves providing human-understandable reasons for model decisions. (6 marks)**
- **There's often a trade-off between interpretability and model performance, as complex models may be less interpretable but potentially more accurate. Models with lots of parameters have larger structure which makes them difficult to understand. (4 marks)**

(10 marks)

(c) Discuss the difference between supervised and unsupervised learning in Data Mining and give an example of when using one or the other.

In supervised learning, the model is trained on labeled data with known outcomes (3 marks), while in unsupervised learning, the model finds patterns in unlabeled data (3 marks). An example of supervised learning is email spam classification (2 marks), and an example of unsupervised learning is customer segmentation based on purchasing behavior (2 marks).

(10 marks)

2. Answer the following three points: (30 marks)

- (a) Explain the apriori algorithm, including what the terms "confidence," "support," and "lift" stand for.

The Apriori algorithm is used for association rule mining in market basket analysis (4 marks). Confidence measures how often a rule is true (2 marks), support measures the frequency of an itemset (2 marks), and lift quantifies the importance of the rule compared to random chance (2 marks).

(10 marks)

- (b) Discuss the difference between filters and wrapper methods including when is best using one or the other.

Filter methods select features based on statistical measures like correlation (3 marks), while wrapper methods use model performance as a criterion for feature selection (3 marks). Filter methods are computationally less expensive and are recommended for high-dimensional data (2 marks). Wrapper methods are computationally expensive but can lead to better model performance (2 marks).

(10 marks)

- (c) Give a short explanation of how Random Forest, K-Nearest Neighbor, Naive Bayes and Support Vector Machine work.

- **Random Forest:** An ensemble learning method that combines multiple decision trees for robust predictions, trained on subsets of data and features. (2.5 marks)
- **K-Nearest Neighbor (KNN):** An instance-based classification algorithm that classifies data points based on the majority class of their k-nearest neighbors, using a user-defined parameter k and a distance metric. (2.5 marks)
- **Naive Bayes:** A probabilistic classifier that assumes feature independence, calculating class probabilities based on feature likelihoods and prior probabilities. (2.5 marks)
- **Support Vector Machine (SVM):** A supervised learning algorithm that finds the optimal hyperplane to maximize the margin between classes, capable of handling binary and multiclass classification tasks through kernel functions. (2.5 marks)

(10 marks)

3. Answer four out of the following five questions: (40 marks)

- (a) Explain how Euclidean Distance, Symbolic Aggregate Approximation and Dynamic Time Warping algorithms work and give an example on where they can be applied.

- **Euclidean Distance** calculates geometric distance, suitable for time series with little misalignment. (2.5 marks)
- **Symbolic Aggregate Approximation (SAX)** converts time series data into a sequence of symbols by discretizing and representing its patterns using a predefined alphabet. Then calculates the distance between the generated strings. (2.5 marks)
- **Dynamic Time Warping (DTW)**, on the other hand, accommodates non-linear alignments, making it ideal for comparing time series with phase shifts, varied time scales, or distortions. (2.5 marks)
- They can be applied for time series clustering either hierarchical or k-means. (2.5 marks)

(10 marks)

- (b) Explain the concepts of Word2Vec and Bag of Words in Text Processing. And an example on where to apply each of them.

- **Word2Vec is a technique for word embedding that represents words as dense vectors. (3 marks)**
- **Word2Vec is used in tasks like text classification, recommendation systems, and text generation. (2 marks)**
- **Bag of Words is a text representation method that treats documents as unordered collections of words, ignoring word order. (3 marks)**
- **BoW is useful for text classification, information retrieval, topic modeling, document clustering, and spelling correction. (2 marks)**

(10 marks)

- (c) Discuss the main steps of the CRISP-DM process.

Main steps for CRISP-DM:

- **Business Understanding:** Define the problem, objectives, and requirements. Develop a clear understanding of the business context (1.5 marks).
- **Data Understanding:** Collect and explore data relevant to the problem. Assess data quality, quantity, and structure (1.5 marks).
- **Data Preparation:** Clean and preprocess the data. Select relevant features and transform the dataset (1.5 marks).
- **Modeling:** Choose appropriate modeling techniques. Train and evaluate models to find the best fit (1.5 marks).
- **Evaluation:** Assess model performance using metrics and cross-validation. Determine if the model meets business objectives (2 marks).
- **Deployment:** Deploy the model into the operational environment. Develop a plan for model monitoring and maintenance (2 marks).

(10 marks)

- (d) Discuss the concept of Reinforcement Learning and its relationship with Markov Decision Processes.

- **Reinforcement Learning (RL)** is a machine learning approach where an agent learns by taking actions in an environment to maximize rewards. It involves the agent's policy, actions, and rewards, aiming to find the best strategy for long-term gain. (5 marks)
- **Markov Decision Processes (MDPs)** provide the mathematical foundation for many RL problems, defining the states, actions, transitions, and rewards. RL can be viewed as solving MDPs, using them to find optimal policies by estimating state values and making decisions to maximize cumulative rewards. MDPs ensure a simplified, memoryless representation of the environment, enabling the agent to learn effectively in sequential decision-making scenarios. (5 marks)

(10 marks)

- (e) Give short definitions for the concepts related to explainability: Accountability, Trustworthiness, Confidence, Causality, and "Fairness and ethical decision".

- **Accountability:** Holding individuals or systems responsible for their actions and decisions, ensuring transparency and traceability in the decision-making process. (2 marks)
- **Trustworthiness:** The quality of being reliable and dependable, indicating users can have confidence in the system's performance and results. (2 marks)
- **Confidence:** The level of certainty or trust in the accuracy and reliability of a system's predictions, often expressed as a probability or confidence score. (2 marks)
- **Causality:** The relationship between cause and effect, exploring how one event or variable influences another, crucial for AI explainability. (2 marks)
- **Fairness and Ethical Decision:** Ensuring that AI decisions align with ethical principles, do not discriminate, and promote equitable outcomes in accordance with societal values. (2 marks)

(10 marks)